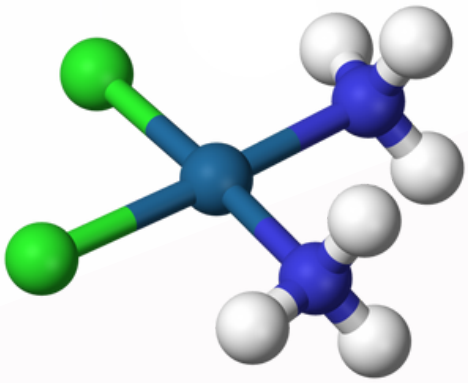
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Inorganic Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Coordination Chemistry

English Medium

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CO-ORDINATION CHEMISTRY

☐ INTRODUCTION :

(a) The concept of co-ordination compounds arises from the complex formation tendency of transition elements.

(b) These compounds play a vital role in our lives, as chlorophyll of plants, vitamin B_{12} and haemoglobin of animal blood are the co-ordination compounds of Mg, Co and Fe respectively.

(c) The co-ordination compounds play important role in analytical chemistry, polymerisation reactions, metallurgy and refining of metals, photography, water purification etc.

(d) Co-ordination compounds also find many applications in electroplating, textile dyeing and medicinal chemistry.

☐ ADDITION COMPOUNDS :

When solutions containing two or more salts in simple molecular proportion are evaporated, crystals of new compound separate out.

These compounds are called molecular or addition compounds.

Ex. $K_2SO_4 + Al_2(SO_4)_3 + 24\ H_2O \longrightarrow K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

$CuSO_4 + 4NH_3 + H_2O \longrightarrow [Cu(NH_3)_4]SO_4 \cdot H_2O$

These addition compounds can be divided into two classes :

(A) DOUBLE SALTS :

Those which lose their identity in solution

In solutions these compounds break down into simpler ions. Such addition compounds which lose their identity in solutions are called **double salts** .

Example :

Potash alum $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$ when dissolved in water breaks down into K^+, SO_4^{2-}, Al^{+3} ions and therefore is an example of **double salt**.

(B) COORDINATION COMPOUNDS :

Those which retain their identity in solution.

In aqueous solution, these addition compounds do not furnish all simple ions but instead give more complex ions having complicated structure .

Example :

Potassium ferrocyanide $K_4[Fe(CN)_6]$ does not furnish simple K^+, Fe^{2+} and CN^- ions but gives K^+ ions and complex ferrocyanide ions, $[Fe(CN)_6]^{4-}$. These types of compounds are called **complex compounds or co-ordination compounds**.

On the basis of stability of complex ion, complex ions are further divided as follows –

(i) Perfect complexes : The compounds in which complex ion is fairly stable and further dissociation or feebly dissociation is not possible in solution state.

Ex. $K_4[Fe(CN)_6] \longrightarrow 4K^+ + [Fe(CN)_6]^{4-}$

$\downarrow$

$Fe^{2+} + 6CN^-$ (Feebly dissociated)

The ferrocyanide ion $[Fe(CN)_6]^{4-}$ is so insignificantly dissociated that it can be considered as practically undissociated and does not give the qualitative test of Fe^{2+} or CN^- ions..

(ii) Imperfect complexes : Those complexes in which complex ion is less stable and is reversibly dissociated to give enough simple ions and thus respond to their usual qualitative test.

Ex. $K_2[Cd(CN)_4] \longrightarrow 2K^+ + [Cd(CN)_4]^{2-}$

$\Downarrow\Uparrow$

$Cd^{2+} + 4CN^-$ (appreciably dissociated)

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☐ DEFINITIONS OF TERMS USED IN CO-ORDINATION COMPOUNDS

(a) **Co-ordination or complex compound :** Co-ordination compounds are those molecular compounds which retain their identity even when dissolved in water or any other solvent and their properties are completely different from those of the constituent ions.

(b) **Central ion :** The cation to which one or more neutral molecules or ions are attached is called the atom / ion. Since, the central ion acts as an acceptor and thus has to accommodate electron pairs donated by the donor atoms of neutral molecules or ions, it must have empty orbitals of appropriate energy.

(c) **Complex ion :** A complex ion may be defined as an electrically charged radical which is formed by the combination of a simple cation with one/more neutral molecules or one/more simple.

(d) **Co-ordination number :** The total number of co-ordinate covalent bond formed by central metal in complex is called the co-ordination number of the central metal ion .

Some common co-ordination number of important metals are as given below.

Metal	Coordination Number	Metal	Coordination Number
Cu^{+}	2, 4	Ni^{2+}	4, 6
Ag^{+}	2	Fe^{2+}	4, 6
Au^{+}	2, 4	Fe^{3+}	6
Hg_2^{2+}	2	Co^{2+}	4, 6
Cu^{2+}	4, 6	Co^{3+}	6
Ag^{2+}	4	Al^{3+}	6
Pt^{2+}	4	Sc^{3+}	6
Pd^{2+}	4	Pt^{4+}	6
Mg^{2+}	6	Pd^{4+}	6

Example. Coordination number of the central metal ions in

(i) $[Cu(NH_3)_4]^{2+}$ is four (ii) $[Fe(EDTA)]^{-}$ is six

(e) **Co-ordination sphere :** The part of the complex enclosed in square bracket is known as co-ordination sphere. It is actually combination of central metal and ligands.

(f) **Ligands :**

(a) The ions or neutral molecules which combine with central metal ion to form complex are called ligands.

(b) They act as electron pair donor (i.e. Lewis bases) though certain ligands also accept electron from central metal and such ligands are known as π - acid ligands.

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❖ CLASSIFICATION OF LIGANDS

(A) Based on charge

(i) Neutral ligands : H_2O, NO, CO, C_6H_6 etc.
(ii) Positive ligands : NO^+, $N_2H_5^+$
(iii) Negative ligands : Cl^-, NO_2^-, CN^-, OH^-

(B) Based on denticity

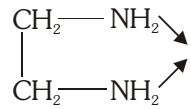
The number of electron pairs donated to central metal by a particular ligand is known as DENTICITY. Depending on number of electron pairs donated, these are classified in following categories.

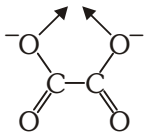
(a) Unidentate/monodentate ligands

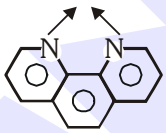
Ligands which donate one pair of electron to the central metal are called unidentate ligands. X^-, CN^-, NO_2^-, NH_3, Pyridine, OH^-, NO_3^-, H_2O, SO_3^{-2}, CO, NO, OH^-, O^{-2}, $(C_6H_5)_3P$ etc.

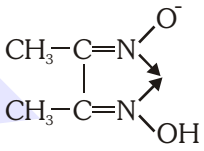
(b) Bidentate ligands

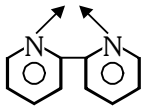
Ligands which have two donor atoms and have the ability to link with central metal ion at two positions are called bidentate ligands.

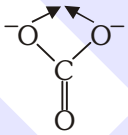

Ethylenediamine (en)

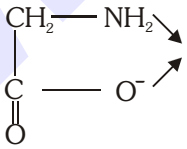

Oxalate (ox)


1, 10-Phenanthroline (O-phen)


Dimethyl glyoxim ion (DMG)


2, 2'-Dipyridyl (Dipy)

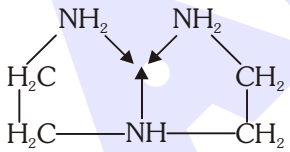

Carbonate

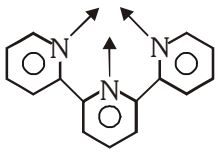

Glycinato (Gly)

(c) Tridentate ligands

The ligands which donate three pairs of electrons to the central metal are called tridentate ligands

Example.

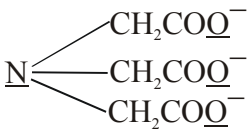

Diethylene triamine (Dien)


2, 2', 2''-Terpyridine (Terpy)

(d) Tetradentate ligands

Those ligands which can donate four electron pairs to the central metal are known as tetradentate ligands,

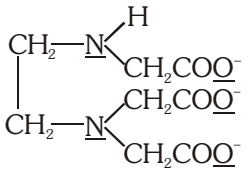
Example : (Underline atoms are donating atom)


Nitrilotriacetato (nta^{3-})

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(e) Pentadentate ligands : Those ligands which can five electron pairs to the central metal are known as pentadentate ligands.

Example : Ethylenediamine triacetate ion. (Underline atoms are donating atom)

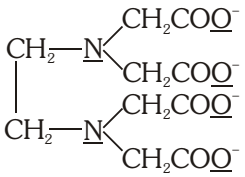


$(EDTA)^{-3}$

Ethylenediaminetriacetate ion

(f) Hexadentate ligands : Those ligands which can donate six electron pairs to the central metal are known as hexadentate ligands.

Example : (Underline atoms are donating atom)

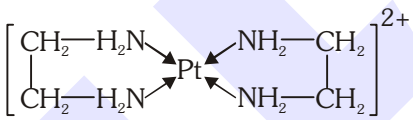


Ethylenediaminetetraacetate ion $(EDTA)^{-4}$

(g) Chelating ligands

Polydentate ligands whose structures permit the attachment of two or more donor site to metal ion simultaneously, thus resulting in cyclic structure are called chelating ligands and compound formed is known as **chelate compound**.

Example :



(h) Ambidentate ligands

Ligands which can ligate through two different atoms present in it are called ambidentate ligands. At a time only one atom can donate.

$\begin{bmatrix} CN^- \\ NC^- \end{bmatrix}$ $\begin{bmatrix} NO_2^- \\ ONO^- \end{bmatrix}$ $\begin{bmatrix} SCN^- \\ NCS^- \end{bmatrix}$ $\begin{bmatrix} CNO^- \\ NCO^- \end{bmatrix}$ $\begin{bmatrix} S_2O_3^{2-} \\ OSO_2S^{-2} \end{bmatrix}$ $\begin{bmatrix} SeCN^- \\ NCSe^- \end{bmatrix}$

Example :

CN^- can coordinate through either the nitrogen or the carbon atom to central metal ion.

(i) Flexidentate ligands

Ligands which sometimes do not use all the donor sites to get coordinated with central metal ion are known as flexidentate ligands.

Ex. SO_4^{2-} , CO_3^{2-} etc.

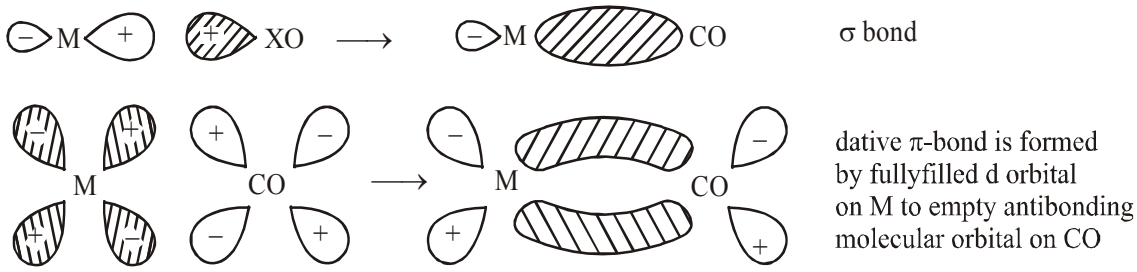
(C) Based upon bonding interaction between the ligand and the central atom.

(i) Classical or simple ligand : These ligand only donate the lone pair of electrons to the central atom.
eg. : O^{2-}, OH^-, F^- etc.

(ii) Non classical or π-acid or π-acceptor ligand : These ligand not only donate the lone pair of electrons to central metal but also accept the electron cloud from central atom
eg. : CO, CN^-, NO^+, PF_3, PR_3 etc.

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BONDING IN METAL CARBONYL : **Ex.** $[Fe(CO)_5]$; $[Ni(CO)_4]$; $[Cr(CO)_6]$



Schematic diagram of orbital overlaps in metal carbonyls.

(a) The metal-carbon bond in metal carbonyls may be represented as the donation of an electron pair from carbon to vacant orbital of metal & form σ bond (M ⟵ CO).

(b) A second bond is formed by back bonding sometimes called dative π-bonding. This is arises from side ways overlap of a full orbital on the metal with the empty antibonding π^*p_y/π^*p_z (if x-axis is molecular axis) molecular orbital of the carbon monooxide, thus forming a π M $\xrightarrow{d\pi-p\pi}$ CO. bond (dπ-pπ back bond).

(c) The filling or partial filling, of the antibonding orbital on CO reduces the bond order of C–O bond from the triple bond in CO towards a double bond. This shown by the increase in C–O bond length from 1.128Å in CO to about 1.15 Å in many carbonyls.

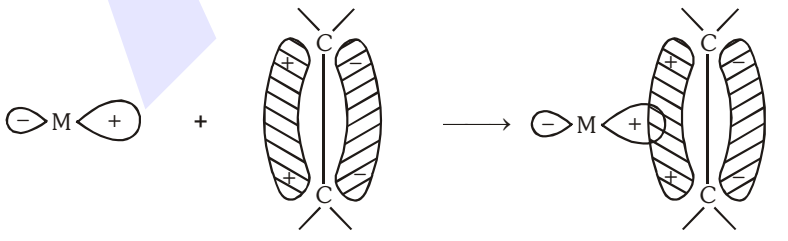
(d) Since CO accept the back donated electrons from the metal atom in to its vacant π* orbital, CO is called π-acid or π-acceptor ligand or π-bonding ligand. Other such π-acid ligands are– CN^-, RCN, $\overset{+}{N}O$.

Note:- π-acid ligands like PF_3, PPh_3 $AsCl_3$ etc. accept the back donated electrons from the metal atom in to its vacant d-orbital of central atom.

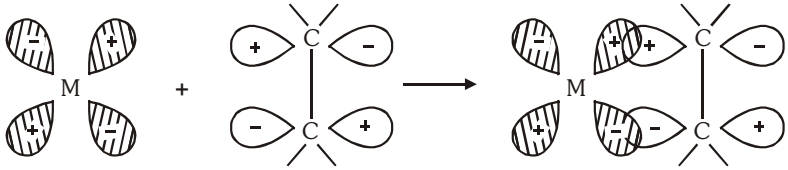
Bonding in π-bonded organo metallic compound. Like zeises salt K $[Pt\ Cl_3(\pi\text{-}C_2H_4)]$

The bonding of alkenes to a transition metal to form complexes has two components. First, the π-electron density of the alkene overlaps with a σ-type vacant orbital or the metal atom.

Second is the π back bond formed by the flow of electron density from a filled d-orbital on the metal into the vacant π^*-antibonding molecular orbital on the alkene molecule as shown below :



σ-overlap



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☐ IUPAC NOMENCLATURE OF COORDINATION COMPOUNDS

The main rules of naming of complexes are –

(a) Like simple salts, the positive part of the coordination compound is named first.

Ex. $K_4[Fe(CN)_6]$ the naming of this complex starts with potassium.

$[Cr(NH_3)_6]Cl_3$ the naming of this complex starts with name of complex ion.

(b) Naming of coordination sphere :- The names of ligands along with their numerical prefixes (to represent their no) are written first, followed by the name of central metal.

(c) The ligands can be neutral, anionic or cationic.

(i) The neutral ligands are named as the molecule **Ex**. C_5H_5N pyridine, $(C_6H_5)_3P$ Triphenyl phosphine.

H_2N — CH_2—CH_2—NH_2 ethylene diamine.

The neutral ligands which are not named as the molecule are CO carbonyl, NO nitrosyl, H_2O Aqua, NH_3 ammine.

(ii) Anionic ligands ending with 'ide' are named by replacing the 'ide' with suffix 'O'.

Symbol	Name as ligand	Symbol	Name as ligand
Cl^-	Chloro/Chlorido	N^{3-}	Nitrido
Br^-	Bromo/Bromido	O_2^{2-}	Peroxo/Peroxido
CN^-	Cyano/Cyanido	O_2H^-	Perhydroxo/Perhydroxido
O^{2-}	Oxo/Oxido	S^{2-}	Sulphido
OH^-	Hydroxo/Hydroxido	NH^{2-}	Imido
H^-	Hydrido/Hydrido	NH_2^-	Amido

Ligands whose names end in 'ite' or 'ate' become 'ito' i.e., by replacing the ending 'e' with 'o' as follows.

Symbol	Name as ligand	Symbol	Name as ligand
CO_3^{2-}	Carbonato	SO_3^{2-}	Sulphito
$C_2O_4^{2-}$	Oxalato	CH_3COO^-	Acetato
SO_4^{2-}	Sulphato	ONO^-	(bonded through oxygen) nitrito
NO_3^-	Nitrato	NO_2^-	(bonded through nitrogen) nitro
$S_2O_3^{-2}$	Thiosulphato		

(iii) Positive ligands naming ends in 'ium' NH_2—NH_3^+ Hydrazinium, NO^+ nitrosonium/nitrosylium.

(d) If ligands are present more than once, then their number is indicated by prefixes like di, tri, tetra etc.

(e) If words like di, tri, tetra are already used in the naming of ligand, or if it is polydented ligand or organic ligand, the prefixes bis–, tris– tetrakis–, pentakis– etc. are used to specify their number.

Example : $[Pt(en)_2Cl_2]Cl_2$: Dichlorobis(ethylenediamine)platinum(IV) chloride.

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(f) When more than one type of ligands are present in the complex, then the ligands are named in the alphabetical order.

(g) After naming of ligands the central metal ion is to be named followed by its oxidation state in Roman numbers in brackets. (as per IUPAC)

If the complex is neutral or provides a cationic complex ion, then the central metal ion is to be named as it is.

If the complex provides anionic complex ion then the name of central metal ion ends with 'ate'

Example : $(NH_4)_2[CuCl_4]$: Ammonium tetrachloridocuprate(II)

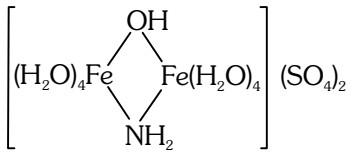
(h) After the naming of central metal ion, anion which is in the outer sphere is to be named.

The naming of some of the complexes is done as follows – (as per IUPAC)

	Complex Compounds	**IUPAC Name**
(i)	**$K_4[Fe(CN)_6]$** (anionic complex) so suffix 'ate' is added with metal name	Potassium hexacyanoferrate(II)
(ii)	**$K_2[Pt\ Cl_6]$**	Potassium hexachloridoplatinate(IV)
(iii)	**$[Co\ (NH_3)_6]\ Cl_3$** (Cationic complex) so metal is without any suffix	Hexamminecobalt(III) chloride
(iv)	**$[Cr(H_2O)_4Cl_2]\ Cl$**	Tetraaquadichloridochromium(III) chloride
(v)	**$[Pt(NH_3)_2Cl_4]$**	Diamminetetrachloridoplatinum(IV)
(vi)	**$[Co(NH_3)_3\ Cl_3]$** (Neutral complex) So no suffix is used with metal ion	Triamminetrichloridocobalt(III)
(vii)	**$K_3[Co(NO_2)_6]$**	Potassium hexanitrocobaltate(III)
(viii)	**$Na_3[Fe(CN)_5\ NO]$**	Sodium pentacyanonitrosylferrate(II)
(ix)	**$[NiCl_4]^{-2}$**	Tetrachloridonickelate(II) ion
(x)	**$[Ru(NH_3)_5Cl]^{+2}$**	Pentamminechloridoruthenium(III) ion
(xi)	**$[Fe(en)_3]Cl_3$**	Tris(ethylenediamine)iron(III) chloride
(xii)	**$[Ni\ (Gly)_2]$**	Bis(glycinato)nickel(II)

(i) If a complex ion has two metal atoms then it is termed polynuclear. The ligand which connects the two metal ions is called as **Bridging ligand or Bridge group**.

A prefix of Greek letter μ, is repeated before the name of each different kind of bridging group.



Tetraaquairon(III)-μ-amido-μ-hydroxotetraaquairon(III) sulphate

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FORMATION OF CO-ORDINATION COMPOUNDS

It can be explained by number of theories.

(A) Werner's co-ordination theory

(B) Sidwick theory or Effective Atomic Number Theory (EAN)

(C) Valence bond theory

(D) Crystal field theory

(A) WERNER'S CO-ORDINATION THEORY :

Werner's co-ordination theory was the first attempt to explain the bonding in co-ordination compounds. The main postulates of this theory are :

(a) Metals possesses two types of valencies - Primary valency and secondary valency.

(b) Primary valencies are normally ionisable and are exhibited by a metal in the formation of its simple salts such as $CoCl_3$, $CuSO_4$ and AgCl. In these salts the primary valencies of Co, Cu and Ag are 3, 2, 1 respectively. Primary valencies are referred to as oxidation state of their metal ion.

(c) Secondary valencies are non-ionisable and are exhibited by a metal in the formation of its complex ions such as $[Co(NH_3)_6]^{3+}$, $[Cu(NH_3)_4]^{2+}$ and $[Ag(NH_3)_2]^{+}$. In these complex, the secondary valencies of Co^{3+}, Cu^{2+}, Ag^{+} are 6, 4 and 2 respectively. These are refered to as co-ordination number (C.N.) of the metal cation.

(d) Primary linkages (valencies) are satisfied by negative ions while secondary valencies are satisfied by neutral molecules, negative ions or in some cases positive ions also.

(e) Every metal atom or ion has a fixed number of secondary valencies. In other words, the co-ordination number of the metal atom is usually fixed.

(f) Every metal has tendency to satisfy both its primary and secondary valencies.

(g) The ligands satisfying secondary valency are always directed towards fixed positions in space about the central metal atom or ion. Thus, the co-ordination compounds have a definite geometry. Werner deduced that in $CoCl_3 \cdot 5NH_3$ only two of the three chlorine atoms are ionic and 5 NH_3 and one Cl form co-ordinate bonds to Co^{3+} ion.

Formula of some cobalt complexes.

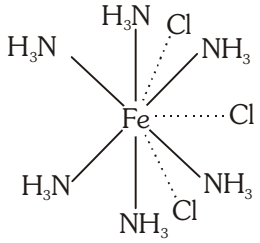
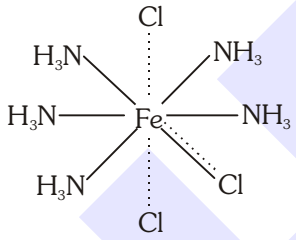
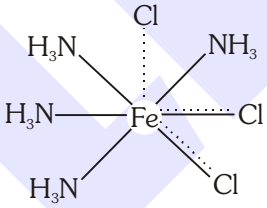
Example :

Old	New	No. of Cl^- Ions precipitated	Total No. of ions
(i) $CoCl_3 \cdot 6\ NH_3$	$[Co(NH_3)_6]Cl_3$	3	4
(ii) $CoCl_5 \cdot 5\ NH_3$	$[Co(NH_3)_5Cl]Cl_2$	2	3
(iii) $CoCl_3 \cdot 4\ NH_3$	$[Co(NH_3)_4Cl_2]Cl$	1	2

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Complex	Modern formula	No. of Cl^- Ions precipitated	Total number of ions
$PtCl_4 . 6NH_3$	$[Pt(NH_3)_6]Cl_4$	4	5
$PtCl_4 . 5NH_3$	$[Pt(NH_3)_5Cl]Cl_3$	3	4
$PtCl_4 . 4NH_3$	$[Pt(NH_3]_4Cl_2]Cl_2$	2	3
$PtCl_4 . 3NH_3$	$[Pt(NH_3)_3Cl_3]Cl$	1	2
$PtCl_4 2NH_3$	$[Pt(NH_3)_2Cl_4]$	0	0 (non-electrolyte)

❖ **WERNER'S REPRESENTATION OF COMPLEXES**

(i)	$Fe(NH_3)_6Cl_3$		$[Fe(NH_3)_6]Cl_3$ Dotted lines indicate primary valency and continuous lines indicate secondary valency of metal ion.
(ii)	$Fe(NH_3)_5Cl_3$		$[Fe(NH_3)_5Cl]Cl_2$ In this complex two 'Cl' groups act as primary valencies and one of the 'Cl' acts as secondary valency also.
(iii)	$Fe(NH_3)_4Cl_3$		$[Fe(NH_3)_4Cl_2]Cl$ In this complex one 'Cl' group act as primary valency and two of the 'Cl' groups act as secondary valencies also.

(B) SIDWICK THEORY OR EFFECTIVE ATOMIC NUMBER CONCEPT (EAN)

Sidwick proposed effective atomic number theory to explain the stability of the complexes. Total number of electrons on central metal including those tranferred from ligands is known as EAN. The EAN generally coincides with the atomic number of next inert gas except in some cases.

EAN can be calculated by the following relation :

EAN = (atomic number of the metal – oxidation state of central metal with sign) + number of electrons gained from the donor atoms of the ligands.

Example Effective atomic number of the metal atom in the following :

(a) $K_3[Cr(C_2O_4)_3]$ is 33 (b) $K_4[Fe(CN)_6]$ is 36

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(C) VALENCE BOND THEORY

The main features of this theory are -

(a) Every metal ion when it forms a complex compound undergoes formation of coordinate covalent bond.

(b) During this bond formation, the metal ion acts as electron pair acceptor. For this the metal ion provides vacant orbitals.

(c) The number of vacant orbitals provided is equal to the coordination number of metal ion.

Example : In the formation of $[Fe(NH_3)_6]^{3+}$, Fe^{+3} ion provides six vacant orbitals.

In $[Cu(NH_3)_4]^{2+}$, Cu^{+2} ion provides four vacant orbitals.

(d) The metal provides vacant orbitals only after the process of hybridisation, thus vacant hybrid orbitals are provided by the metal ion.

(e) The vacant hybrid orbitals of metal ion get overlapped by orbitals of ligands containing lone pair of electrons.

(f) The number of such overlappings is equal to the coordination number of metal ion.

(g) The empty 'd' orbitals involved in hybridisation may be inner (n-1)d or outer "nd" orbitals and accordingly complexes are called as **Inner orbital complexes** and **outer orbital complexes** respectively.

(h) In certain complexes pairing of electrons takes place in ligand field, resulting in decrease in spin only magnetic moment, such complexes are known as Low spin complexes

(i) $\text{Bohr magneton} = \dfrac{eh}{4\pi mc}$

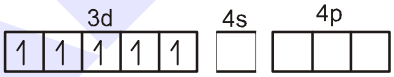
(j) Paramagnetism is represented in the term of spin only magnetic moment.

$\mu = \sqrt{n(n+2)}$ B.M. n = Number of unpaired electron

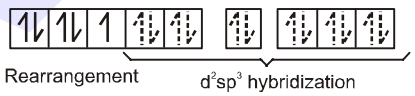
Example $[Fe(CN)_6]^{3-}$ is weakly paramagnetic while $[Fe(CN)_6]^{4-}$ is diamagnetic.

Sol. $[Fe(CN)_6]^{3-}$ involves d^2sp^3 hybridization.

Fe^{3+}, [Ar] $3d^5$



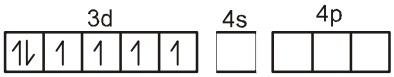
$[Fe(CN)_6]^{3-}$



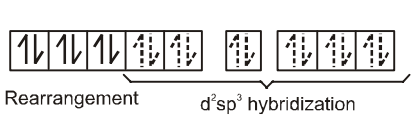
One d-orbital is singly occupied, hence it is weakly paramagnetic in nature.

$[Fe(CN)_6]^{4-}$ also involves d^2sp^3 hybridization but it has Fe^{2+} ion as central ion.

Fe^{2+}, [Ar] $3d^6$



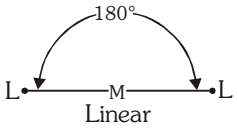
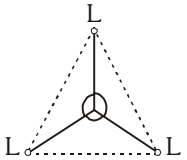
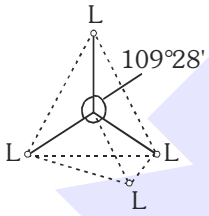
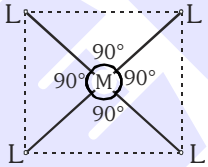
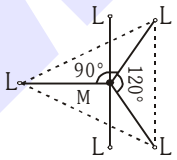
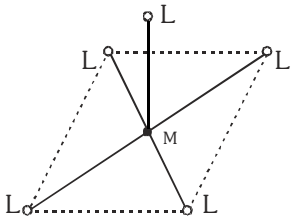
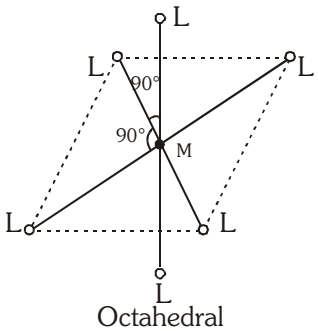
$[Fe(CN)_6]^{4-}$



All electrons are paired, hence it is diamagnetic in nature.

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Some Example :

Coordination Number	Hybridised orbitals	Geometrical shape of the Complex	Examples of Complex
2	sp	 Linear	$[Ag(NH_3)_2]^+$ $[Ag(CN)_2]^-$
3	sp^2		$[HgI_3]^-$
4	sp^3		$[CuCl_4]^{-2}$ $[ZnCl_4]^{-2}$ $[FeCl_4]^-$ $[Ni(CO)_4]$ $[Zn(NH_3)_4]^{+2}$
4	dsp^2 ($d = d_{x2-y2}$)	 Square planar	$[PdCl_4]^{2-}$ $[Ni(CN)_4]^{2-}$ $[Pt(NH_3)_4]^{+2}$ $[Cu(NH_3)_4]^{+2}$ $[PtCl_4]^{2-}$
5	sp^3d ($d = d_{z2}$) or dsp^3 ($d = d_{z2}$)	 Trigonal bipyramidal	$[Fe(CO)_5]$ $[CuCl_5]^{3-}$
5	sp^3d ($d = d_{x2-y2}$) or dsp^3 ($d = d_{x2-y2}$)	 Square pyramidal	$[Ni(CN)_5]^{-3}$
6	d^2sp^3 (inner orbital complex) or sp^3d^2 (outer orbital complex) in both case d-orbitals are d_{z^2} & $d_{x^2-y^2}$.	 Octahedral	$[Cr(NH_3)_6]^{+3}$ $[Ti(H_2O)_6]^{+3}$ $[Fe(CN)_6]^{-3}$ $[Co(NH_3)_6]^{+3}$ $[PtCl_6]^{-2}$, $[CoF_6]^{-3}$

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- **Drawback of valence bond theory :**
  (a) It describes bonding in co-ordination compounds only qualitatively but not account for the relative stabilities for different co-ordination complexes.
  (b) It does not offer any explanation for optical absorption spectra (coloration) of complexes
  (c) It does not describe the detailed magnetic properties of co-ordination compounds.

(D) CRYSTAL FIELD THEORY : The drawbacks of VBT of coordination compounds are, to a considerable extent, removed by the Crystal Field Theory.

The crystal field theory (CFT) is an electrostatic model which considers the metal-ligand bond to be ionic arising purely from electrostatic interaction between the metal ion and the ligand. Ligands are treated as point charges in case of anions or dipoles in case of neutral molecules. The five d-orbitals in an isolated gaseous metal atom/ion have same energy, i.e., they are degenerate. This degeneracy is maintained if a spherically symmetrical field of negative charges surrounds the metal atom/ion. However, when this negative field is due to ligands (either anions or the negative ends of polar molecules like NH_3 and H_2O) in a complex, it becomes asymmetrical and the degeneracy of the d orbitals is lost. It results in splitting of the d orbitals. The pattern of splitting depends upon the nature of the crystal field.

(a) Crystal field splitting in octahedral coordination entities :

In an octahedral coordination entity with six ligands surrounding the metal atom/ion, there will be repulsion between the electrons in d orbitals of metal and the electrons (or negative charges) of the ligands. Such a repulsion is more when the d orbitals of metal are directed towards the ligand than when it is away from the ligand.

Thus, the $d_{x^2-y^2}$ and d_{z^2} orbitals (axial orbitals) which point towards the axis along the direction of the ligand will experience more repulsion and will be raised in energy ; and the d_{xy} , d_{yz} and d_{zx} orbitals (non-axial) orbitals which are directed between the axis will be lowered in energy relative to the average energy in the spherical crystal field.

Thus, the degeneracy of the d orbitals has been removed due to ligand electron-metal electron repulsions in the octahedral complex to yield three orbitals of lower energy, t_{2g} set and two orbitals of higher energy, e_g set. This splitting of the degenerate levels due to the presence of ligands in a definite geometry is termed as crystal field splitting and the energy separation is denoted by Δ_0 (the subscript o is for octahedral). Thus, the energy of the two e_g orbitals will increase by $(3/5)\Delta_0$ and that of the three t_{2g} will decrease by $(2/5)\ \Delta_0$.

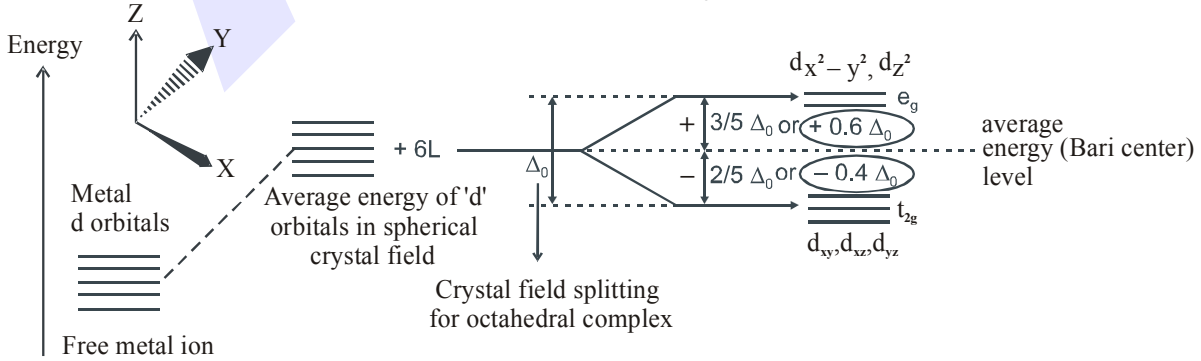


Figure showing crystal field splitting in octahedral complex.

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The crystal field splitting, Δ_0, depends upon the fields produced by the ligand and charge on the metal ion. Some ligands are able to produce strong fields in which case, the splitting will be large whereas others produce weak fields and consequently result in small splitting of d orbitals. In general, ligands can be arranged in a series in the order of increasing field strength as given below :

$I^- < Br^- < SCN^- < Cl^- < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < CN^- < CO$

Note : In SCN^-, S is donating atom and in NCS^-, N is donating atom.

Such a series is termed as spectrochemical series. It is an experimentally determined series based on the absorption of light by complexes with different ligands. For d^4 configuration, the fourth electron will singly occupy e_g orbital (according to Hund's rule) or will undergo pairing in t_{2g} orbital, which of these possibilities occurs, depends on the relative magnitude of the crystal field splitting, Δ_0 and the pairing energy, P (P represents the energy required for electron pairing in a single orbital). The two possibilites are :

(i) If $\Delta_0 < P$, the fourth electron enters in one of the e_g orbitals giving the configuration $t_{2g}^3 e_g^1$. Ligands for which $\Delta_0 < P$ are known as weak field ligands and form high spin complexes.

(ii) If $\Delta_0 > P$, it becomes more energetically favourable for the fourth electron to occupy a t_{2g} orbital with configuration $t_{2g}^4 e_g^0$. Ligands which produce this effect are known as strong field ligands and form low spin complexes.

Crystal Field stabilising energy in Octahedral field :

Formula : $CFSE = [-0.4\ n_{t_{2g}} + 0.6\ n_{e_g}]\ \Delta_0 + xP$.

Where $n_{t_{2g}}$ & n_{e_g} are number of electron(s) in t_{2g} & e_g orbitals respectively and Δ_0 crystal field splitting energy for octahedral complex. "x" represents the number of electron pairs and P is mean pairing energy.

(b) Crystal field splitting in tetrahedral coordination entities :

In tetrahedral coordination entity formation, the d orbital splitting is inverted and is smaller as compared to the octahedral field splitting. For the same metal, the same ligands and metal-ligand distances, it can be shown that $\Delta_t = (4/9)\Delta_0$. This may attributes to the following two reasons.

(i) There are only four ligands instead of six, so the ligand field is only two thirds the size ; as the ligand field spliting is also the two thirds the size and (ii) the direction of the orbitals does not concide with the direction of the ligands. This reduces the crystal field spliting by roughly further two third. So $\Delta_t = \frac{2}{3} \times \frac{2}{3} = \frac{4}{9}\Delta_o$.

Consequently, the orbital splitting energies are not sufficiently large for forcing pairing and, therefore, low spin configurations are rarely observed.

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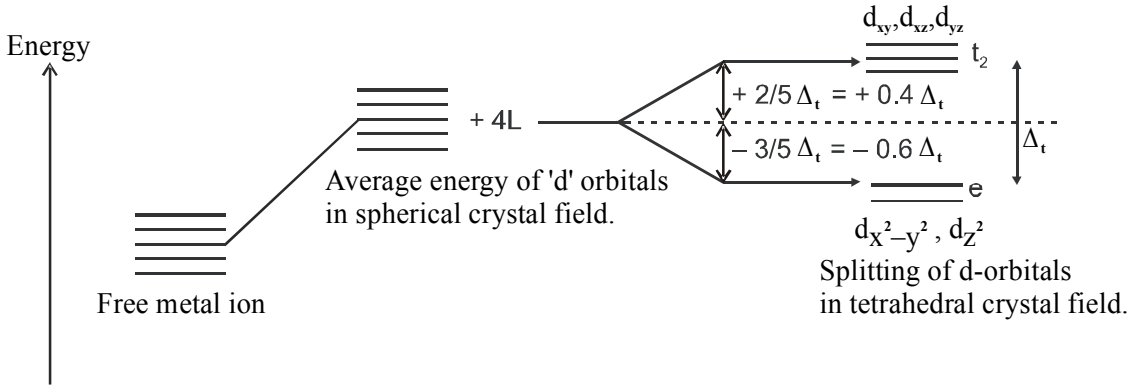


Figure showing crystal field splitting in tetrahedral complex.

Since $\Delta_t < \Delta_o$ crystal field spliting favours the formation of octahedral complexes.

Crystal Field stabilising energy in Tetrahedral field :

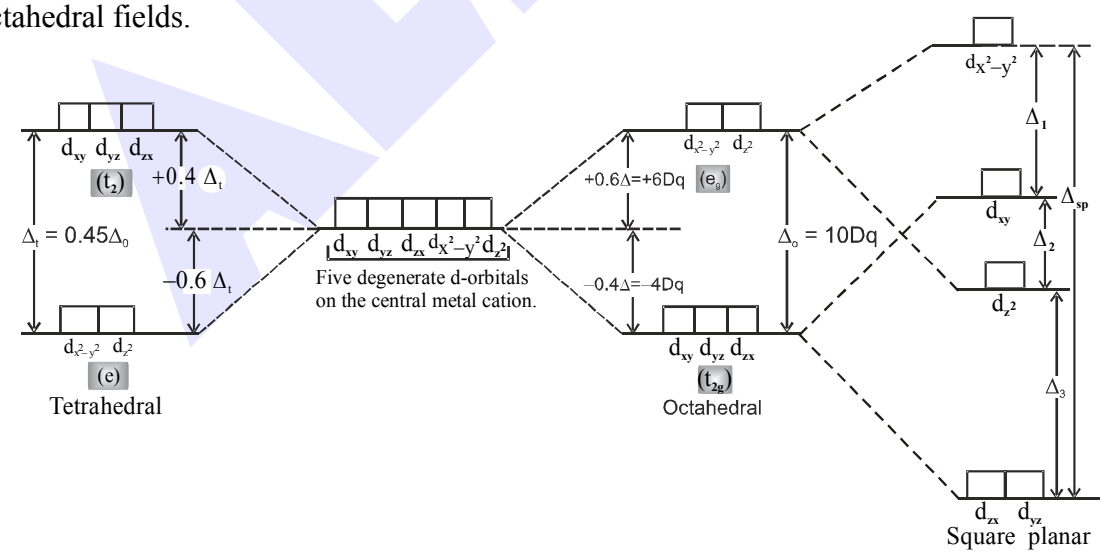
Formula : CFSE = $[-0.6\ n_e + 0.4\ n_{t_2}]\ \Delta_t + xP$.

where n_{t_2} & n_e are number of electron(s) in t_2 & e orbitals respectively and Δ_t crystal field splitting energy for tetrahedral complex. "x" represents the number of electron pairs and P is mean pairing energy.

(c) Crystal field splitting in square planar co-ordination entities :

The square planar arrangement of ligands may be considered to be one derived from the octahedral field by removing two trans-ligands located along the Z-axis. In the process, the e_g and t_{2g} sets of orbitals is lifted i.e., these orbitals will no longer be degenerate.

The four ligands in square planar arrangement around the central metal ion are shown in Fig. As the ligands approach through the x and y axis, they would have greatest influence on $d_{x^2-y^2}$ orbital, so the energy of this orbital, will be raised most. The d_{xy} orbital, lying in the same plane, but between the ligands will also have a greater energy though the effect will be less than that on the $d_{x^2-y^2}$ orbitals. On the other hand, due to absence of ligands along Z-axis, the d_{z^2} orbital becomes stable and has energy lower than that of d_{xy} orbital. Similarly d_{yz} and d_{xz} become more stable. The energy level diagram may be represented as shown in figure along with tetrahedral and octahedral fields.



The value of Δ_{sp} has been found larger than Δ_o because of the reason that d_{xz} and d_{yz} orbitals interact with only two ligands in the square planar complexes, while in octahedral complexes the interaction takes place only with four ligands. Δ_{sp} has been found equal to $1.3\Delta_o$. Thus.

$\Delta_{sp} = (\Delta_1 + \Delta_2 + \Delta_3) > \Delta_o$ and $\Delta_{sp} = 1.3\ \Delta_o$.

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(E). STABILITY OF COORDINATION COMPOUNDS :

The stability of a coordination compound $[ML_n]$ is measured in terms of the stability constant (equilibrium constant) given by the expression,

$$\beta_n = [ML_n]/[M(H_2O)_n][L]^n$$

for the overall reaction :

$$M(H_2O)_n + nL \rightleftharpoons ML_n + nH_2O$$

By convention, the water displaced is ignored, as its concentration remains essentially constant. The above overall reaction takes place in steps, with a stability (formation) constant, K_1, K_2, K_3, ... K_n for each step as represented below :

$$M(H_2O)_n + L \rightleftharpoons ML(H_2O)_{n-1} + H_2O$$

$$K_1 = [ML(H_2O)_{n-1}] / \{[M(H_2O)_n][L]\}$$

$$ML_{n-1}(H_2O) + L \rightleftharpoons ML_n + H_2O$$

$$K_n = [ML_n] / \{[ML_{n-1}(H_2O)][L]\}$$

$$M(H_2O)_n + nL \rightleftharpoons ML_n + nH_2O$$

$$\beta_n = K_1 \times K_2 \times K_3 \times ..... \times K_n$$

β_n, the stability constant, is related to thermodynamic stability when the system has reached equilibrium. Most of the measurements have been made from aqueous solutions, which implies that the complex is formed by the ligand displacing, water from the aqua complex of the metal ion. Ignoring the charge and taking L as an unidentate ligand, the stepwise formation of the complex is represented as shown above.

K_1, K_2, K_3 ..... K_n representing the stepwise stability (or formation) constants.

The above is thermodynamic stability criteria, there can be another kind of stability called kinetic stability, which measures the rate of ligand replacement.

(F). FACTORS INFLUENCING THE MAGNITUDE OF C.F.S.E. :

1. **Different charges on the cation of the same metal :** The cation with a higher oxidation state has a larger value of CFSE than that with lower oxidation state e.g.,

   $$[Fe(H_2O)_6]^{3+} > [Fe(H_2O)_6]^{2+}$$

2. **Same charges on the cation but the number of d-electrons is different :** The metal cation the magnitude of CFSE with the increase of the number of d-electrons, e.g.,

   $$[Co(H_2O)_6]^{2+} < [Ni(H_2O)_6]^{2+}$$

3. **Quantum number (n) of the d-orbitals of the central metal ion :** As 'n' increase CFSE increases.

   $$[Co(NH_3)_6]^{3+} < [Rh(NH_3)_6]^{3+} < [Ir(NH_3)_6]^{3+}$$

4. **Types of Hybridisation :**

   $$\Delta_t = \frac{4}{9}\Delta_0$$

5. **Presence of cheleting ligand increases CFSE :**

   $$[Fe(Ox)_3]^{3-} > [Fe(SCN)_6]^{3-}$$

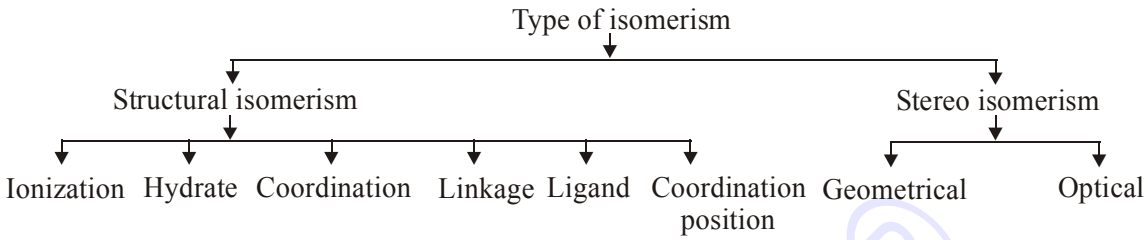
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ISOMERISM IN COMPLEXES

(a) Compounds which have the same molecular formula, but differ in their properties due to the difference in structure are called as **Isomers**.

(b) Isomerism is commonly considered, to be the characteristic of only organic compounds, it is also found although less frequently among inorganic substances.

CLASSIFICATION OF ISOMERISM



(A) Structural isomerism

It arises due to the difference in the type of chemical linkages and distribution of ligands within and outside the coordination sphere.

(a) **Ionisation isomerism**

The type of isomerism which is due to the exchange of groups or ions between the coordination sphere and the ionisation sphere.

Example. (i) $Co(NH_3)_4Br_2SO_4$ can be represented as
$[Co(NH_3)_4Br_2]SO_4$ (red violet) and $[Co(NH_3)_4SO_4]Br_2$ (red)
These complexes give sulphate ion and bromide ion respectively

(ii) $[Pt(NH_3)_4Cl_2]Br_2$ and $[Pt(NH_3)_4Br_2]Cl_2$

(iii) $[Co(NH_3)_4(NO_3)_2]SO_4$ and $[Co(NH_3)_4SO_4](NO_3)_2$

(b) **Hydrate isomerism**

This type of isomerism is due to presence of different number of water molecules inside a coordination sphere.

Example. $Cr(H_2O)_6Cl_3$ has four possible structures

(i) $[Cr(H_2O)_6]Cl_3$ violet

(ii) $[Cr(H_2O)_5Cl]Cl_2.H_2O$ green

(iii) $[Cr(H_2O)_4Cl_2]Cl.2H_2O$ dark green.

(iv) $[Cr(H_2O)_3Cl_3].3H_2O$ dark green.

These complexes differ from one another with respect to the number of water molecules acting as ligands. Other hydrate isomers are
$[Co(NH_3)_4(H_2O)Cl]Cl_2$ and $[Co(NH_3)_4Cl_2]Cl.H_2O$

(c) **Linkage isomerism**

(i) This type of isomerism arises due to presence of ambidentate ligands like
NO_2^-, CN^- and SCN^-

(ii) These ligands have two donor atoms but at a time only one atom is directly linked to the central metal atom of the complex.

(iii) Such type of isomers are distinguished by infra red (I.R.) spectroscopy.

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Example. • $[Co(NH_3)_5NO_2]Cl_2$ and $[Co(NH_3)_5ONO]Cl_2$

• In NO_2^- ligand, The coordinating sites are nitrogen (i.e., NO_2^- Nitro ligand) or through oxygen (i.e. ONO Nitrito ligand)

• The nitro isomer is yellow and is stable to acids whereas nitrito isomer is red and is decomposed by acids.

(d) **Coordination isomerism**

(i) This type of isomerism is exhibited when the complex has two complex ions in it - 'cationic and anionic'.

(ii) This type of isomerism is caused by the interchange of ligands between the two complex ions of the same complex.

Example. $[Co(NH_3)_6]\ [Cr(CN)_6]$ and $[Co(NH_3)_6]\ [Cr(C_2O_4)_3]$

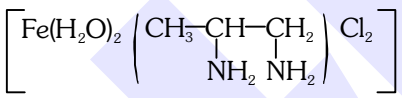
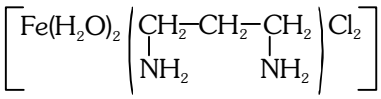
$[\ Cr(NH_3)_6]\ [Co(CN)_6]$ $[Cr(NH_3)_6]\ [Co(C_2O_4)_3]$

(e) **Ligand isomerism**

(i) Ligands with $C_3H_6(NH_2)_2$ have two different structures i.e. 1, 3-diamino propane and 1, 2-diaminopropane(propylene diamine).

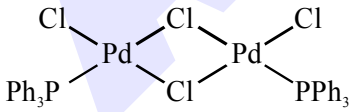
(ii) Those complexes which have same molecular formula, but differ with respect to their ligands are called as **Ligand isomers**.

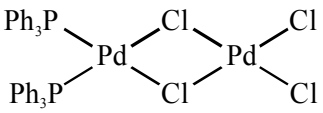
Example. $[Fe(H_2O)_2\ C_3H_6(NH_2)_2Cl_2]$ has two different structures


and


(f) **Co-ordination Position Isomerisation :**

It is shown by polynuclear complexs, due interchange of ligands between the different metal nuclei.



Example.


(g) **Polymerization Isomerism :**

This is not true isomerism because it occurs between compounds having the same empirical formula, but different molecular weights.

Example. $[Pt(NH_3)_2Cl_2]$

$[Pt(NH_3)_4]\ [PtCl_4]$

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(B) Stereo isomerism

They have same molecular formula, same constitution, they differ only with respect to the spatial orientation of ligands in space around the metal ion. The two stereo isomers which are possible Geometrical and Optical.

(a) **Geometrical isomerism**

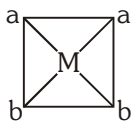
(i) The ligands occupy different positions around the central metal ion.

(ii) When two identical ligands are co-ordinated to the metal ion from same side then it is **cis isomer**. (Latin, cis means same).

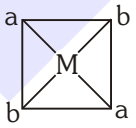
(iii) If the two identical ligands are co-ordinated to the metal ion from opposite side then it is **trans isomer** (in Latin, trans means across).

Geometrical isomers with co-ordination number = 4 (Square planar complexes)

(i) Complexes with general formula, $\mathbf{Ma_2b_2}$ (where both a and b are monodentate) can have cis-and trans isomers.

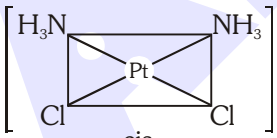


cis-isomer

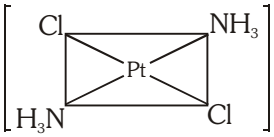


trans-isomer

Example. **$[Pt(NH_3)_2Cl_2]$**

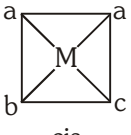


cis
(cis-platin) anti cancer

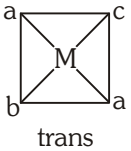


trans

(ii) Complexes with general formula $\mathbf{Ma_2bc}$ can have cis - and trans-isomers.



cis



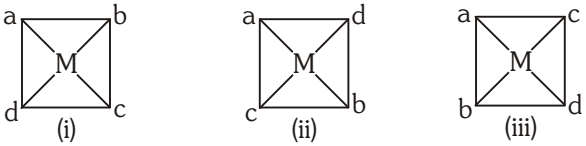
trans

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Example. **$[Pt(NH_3)_2ClBr]$**



(iii) Complexes with general formula, **Mabcd** can have three isomers.



Geometrical isomers with Co-ordination number = 6

(i) Complexes with general formula **Ma_4b_2** can have cis - and trans-isomers.

Example. **$[Fe(NH_3)_4Cl_2]$**



(ii) **Facial and Meridional isomerism (Ma_3b_3)**



Other 6-Coordinated geometrical isomers are

Note :	**General formula**	**Total No. of geometrical isomers**
	Mabcdef	15
	Ma_2bcde	9
	Ma_2b_2cd	6
	$Ma_2b_2c_2$	5
	Ma_3bcd	4
	Ma_3b_2c	3
	Ma_3b_3	2
	Ma_4bc	2
	Ma_4b_2	2
	Ma_5b	Not possible
	Ma_6	Not possible

Here M = central atom a, b, c, d, e, f = Monodentate ligands

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(b) **Optical isomers**

(i) Optically active complexes are those which are nonsuperimposable over the mirror image structure.

(ii) An optically active complex is one which is asymmetric in nature i.e., not divisible into two identical halves.

(iii) The complex which rotates plane polarised light to left hand side is **laevo rotatory i.e. 'ℓ' or '—'** and if the complex rotates the plane polarised light to right hand side then it is **dextro rotatory 'd' or '+'**.

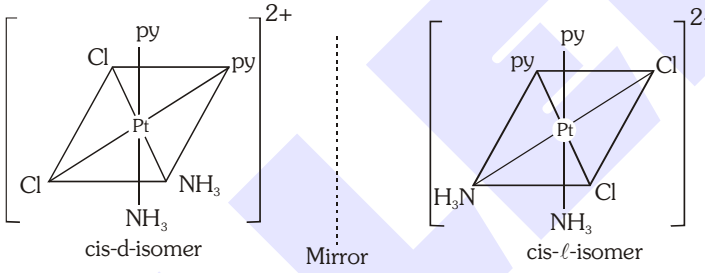
(iv) Thus complexes which have same physical and chemical properties but differ in their action towards plane polarised light are called as **optical isomers**.

(v) The 'd' and 'ℓ' isomers of a compound are called as **Enantiomers or Enantiomorphs**.

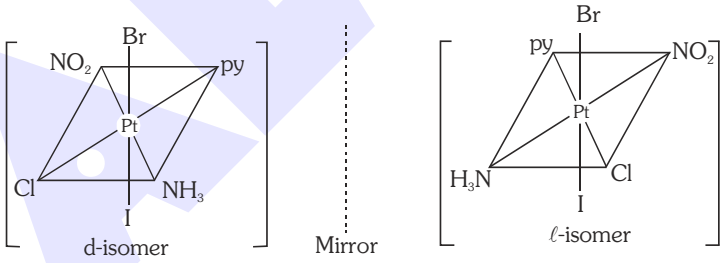
(vi) Optical isomerism is expected in tetrahedral complexes of the type Mabcd.

Optical isomers with Co-ordination number = 6

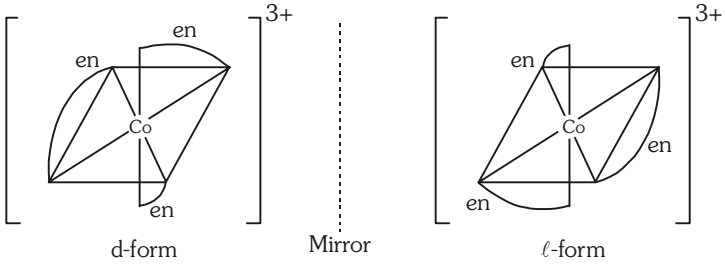
(i) $[Ma_2b_2c_2]^{n+} \rightarrow [Pt(py)_2(NH_3)_2Cl_2]^{2+}$



(ii) **[Mabcedf] → [Pt(py) (NH_3) (NO_2) ClBrI]**



(iii) $[M(AA)_3]^{n+} \rightarrow [Co(en)_3]^{3+}$



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NUMBER OF POSSIBLE ISOMERS FOR SPECIFIC COMPLEXES

Formula	Number of stereoisomers	Pairs of Enantiomers
$M_{a_4b_2}$	2	0
$M_{a_3b_3}$	2	0
M_{a_4bc}	2	0
M_{a_3bcd}	5	1
M_{a_2bcde}	15	6
M_{abcdef}	30	15
$M_{a_2b_2c_2}$	6	1
$M_{a_2b_2cd}$	8	2
$M_{a_3b_2c}$	3	0
M(AA)(BC)de	10	5
M(AB)(AB)cd	11	5
M(AB)(CD)ef	20	10
$M(AB)_3$	4	2

Note : Uppercase letters represent chelating ligands and lowercase letters represent monodentate ligands.

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